Carbohydrates Metabolism: Galactose and Fructose Metabolism

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Objectives:

- At the end of this chapter, it is expected to;
  - Describe the metabolism of fructose, galactose, and other dietary carbohydrates
  - Explain how fructose, galactose, and other dietary carbohydrates feed in to glycolytic pathway
  - Explain the metabolic conditions due to defect in fructose, galactose, and other dietary carbohydrates metabolism
  - Describe the reaction, importance and consequences of polyol pathway

Fructose, galactose, Lactose, Sucrose, Glycogen and starch feed into the glycolytic pathway

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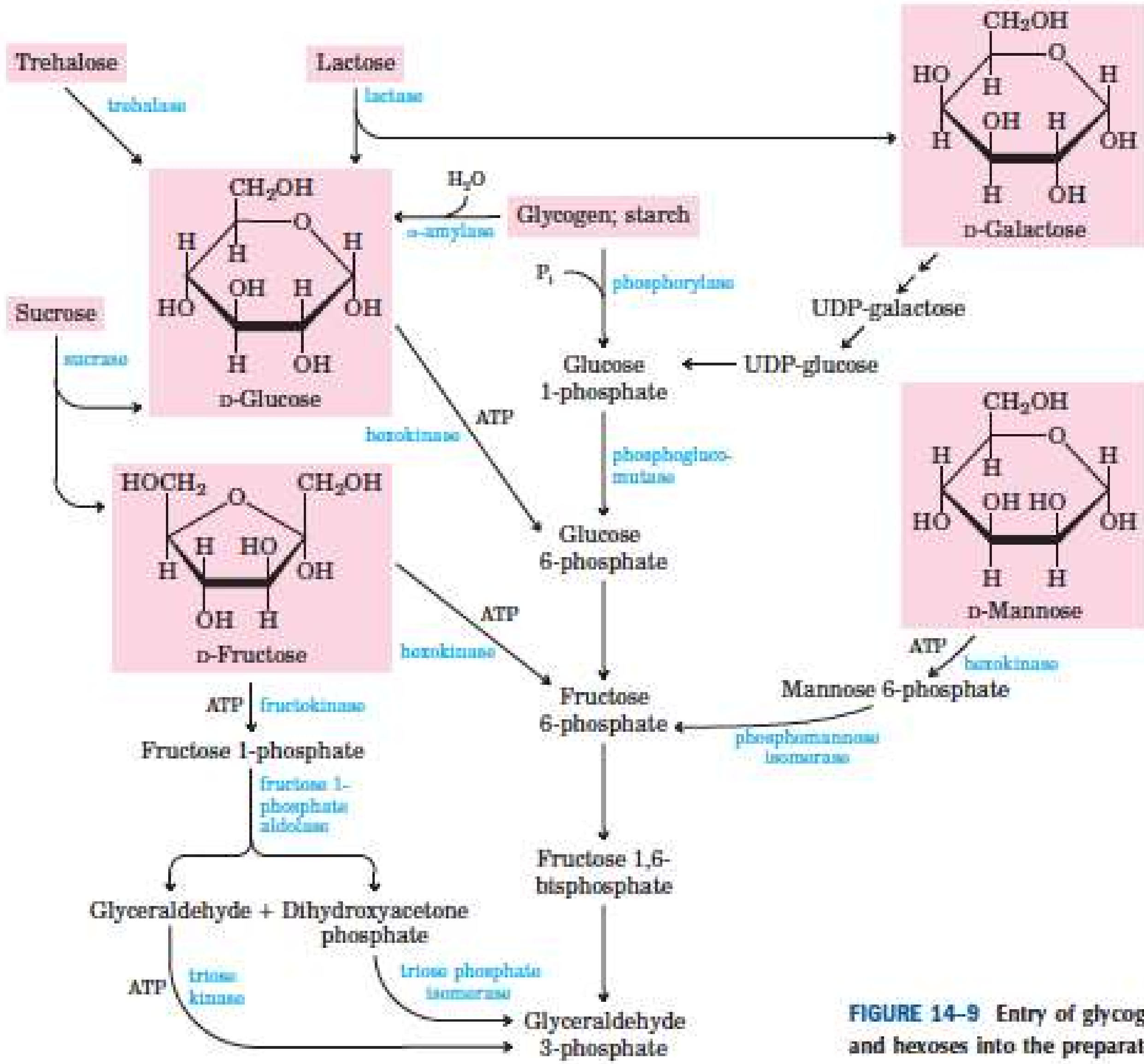
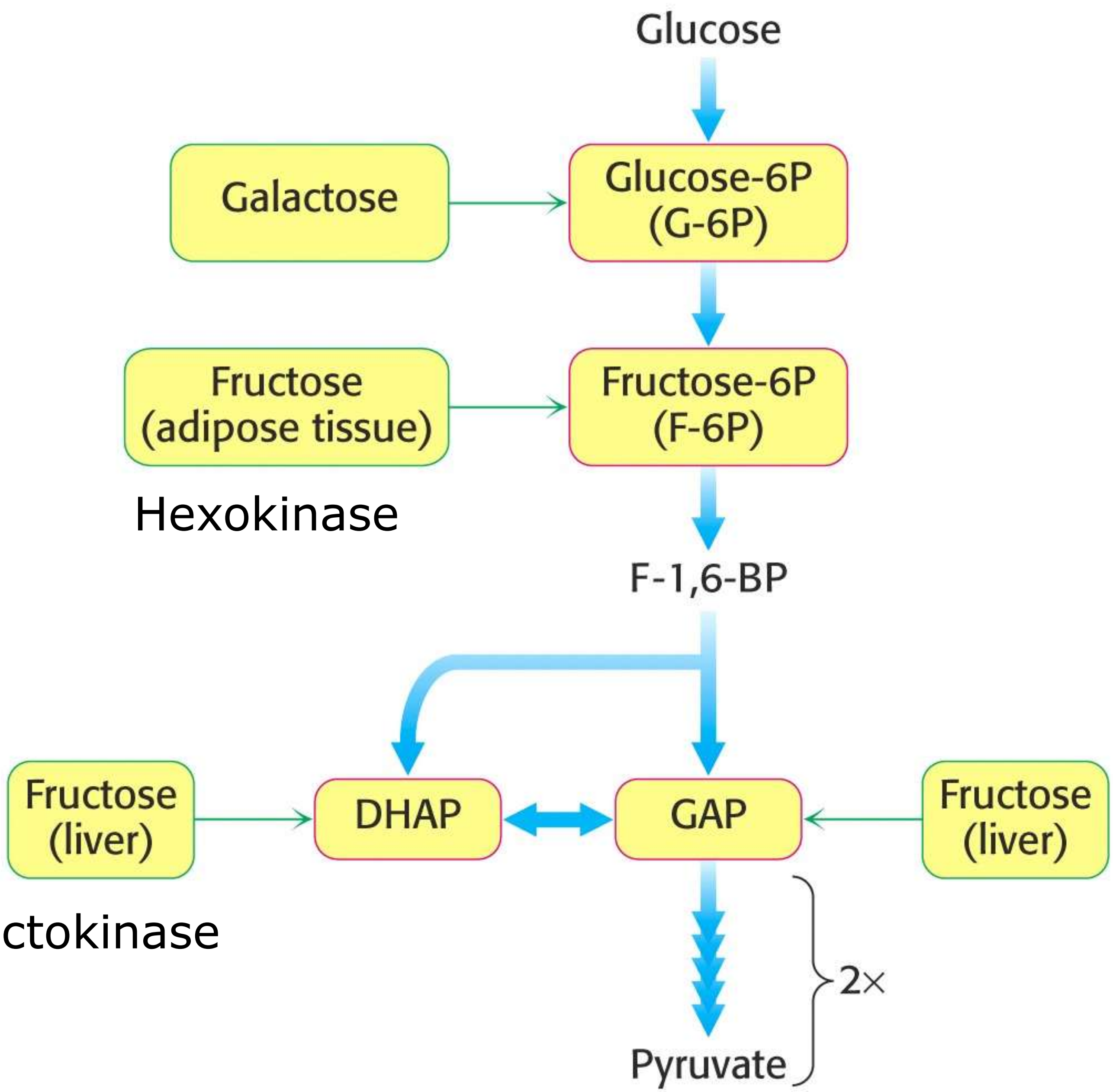


FIGURE 14–9 Entry of glycogen, starch, disaccharides, and hexoses into the preparatory stage of glycolysis.



Hexokinase

Fructokinase

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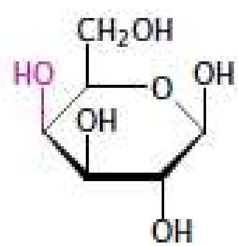
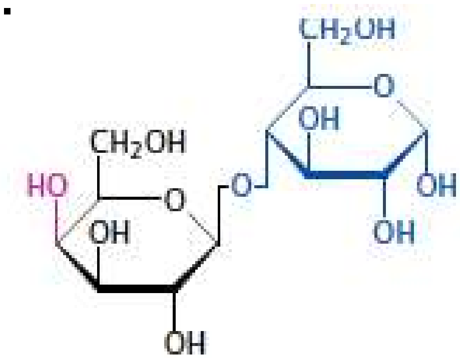
Lactose and galactose

- Lactose - a disaccharide of glucose and galactose
- The major carbohydrate contained in milk.
- Cleaved at the brush border of the small intestine;
  - And the monosaccharide fragments are absorbed and passed along to the liver.
  - The enzyme that accomplishes the cleavage is lactase or, more precisely, β-galactosidase.



β-D-galactose

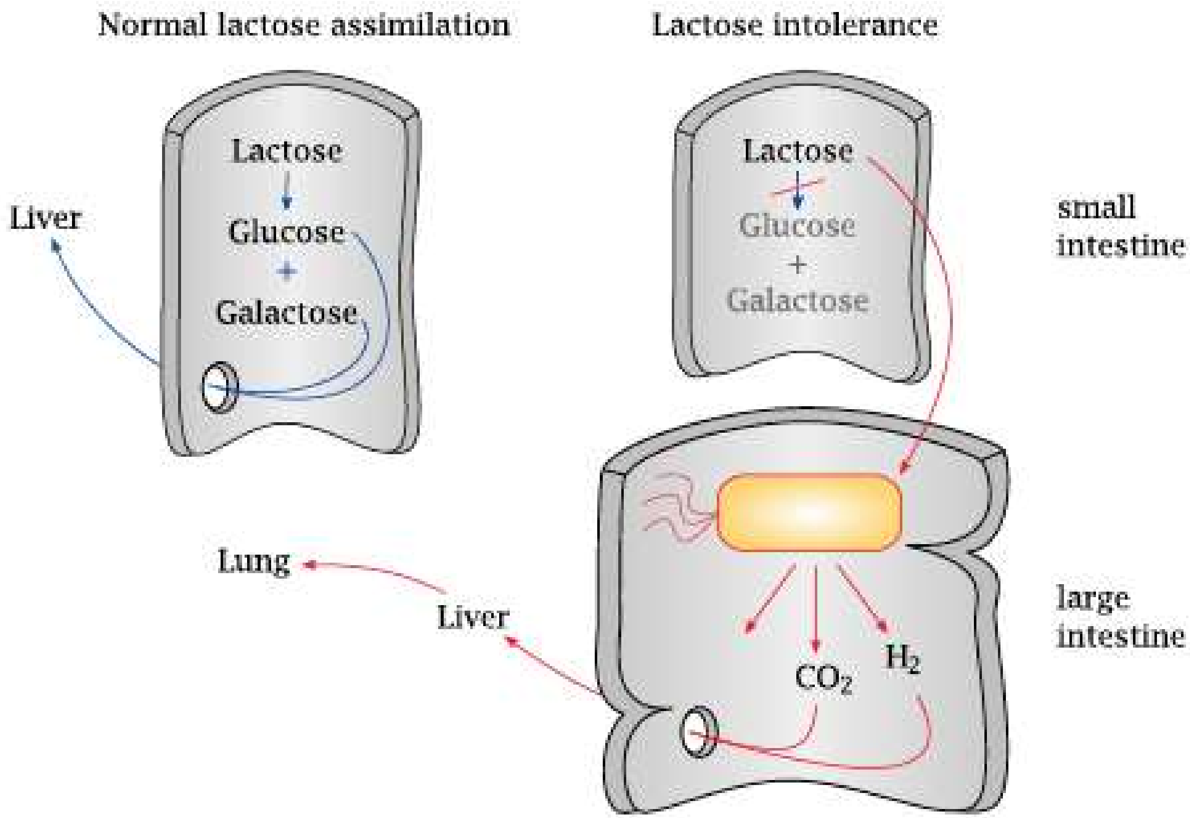


β-D-galactosyl-(1→4)-D-glucoside (lactose)

Lactose intolerance

- A deficiency of the lactase enzyme in the small intestine
- Found frequently in people of East Asian descent who are past their infant age
- Lactose not cleaved in small intestine – moved down to large intestine where get metabolized by bacteria's (E.Coli).
- Mixed acid fermentation ➔ Formic acid (HCOOH), which is then cleaved by formic acid lyase to H_2 and CO_2
- The aberrant fermentation and gas formation leads to abdominal discomfort and diarrhea

Lactose intolerance



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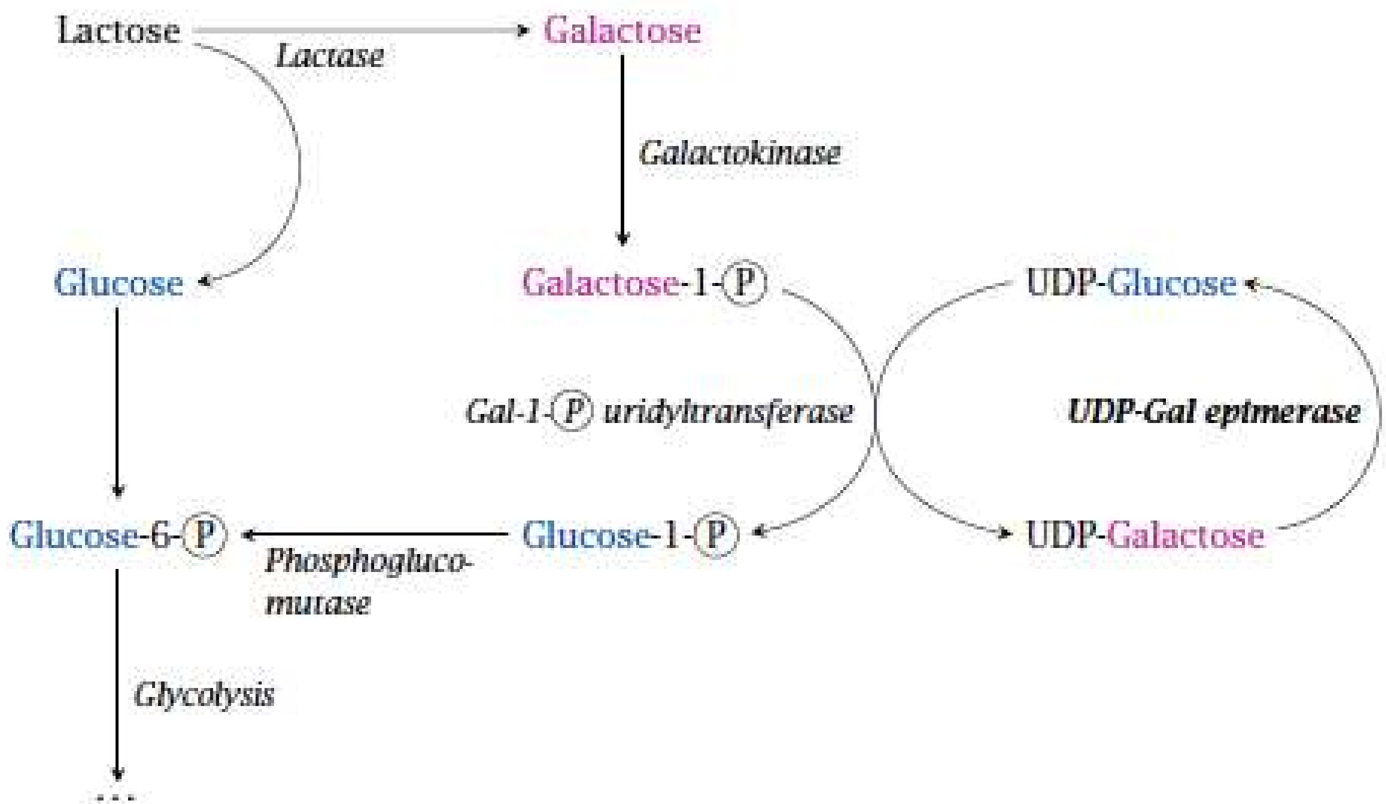
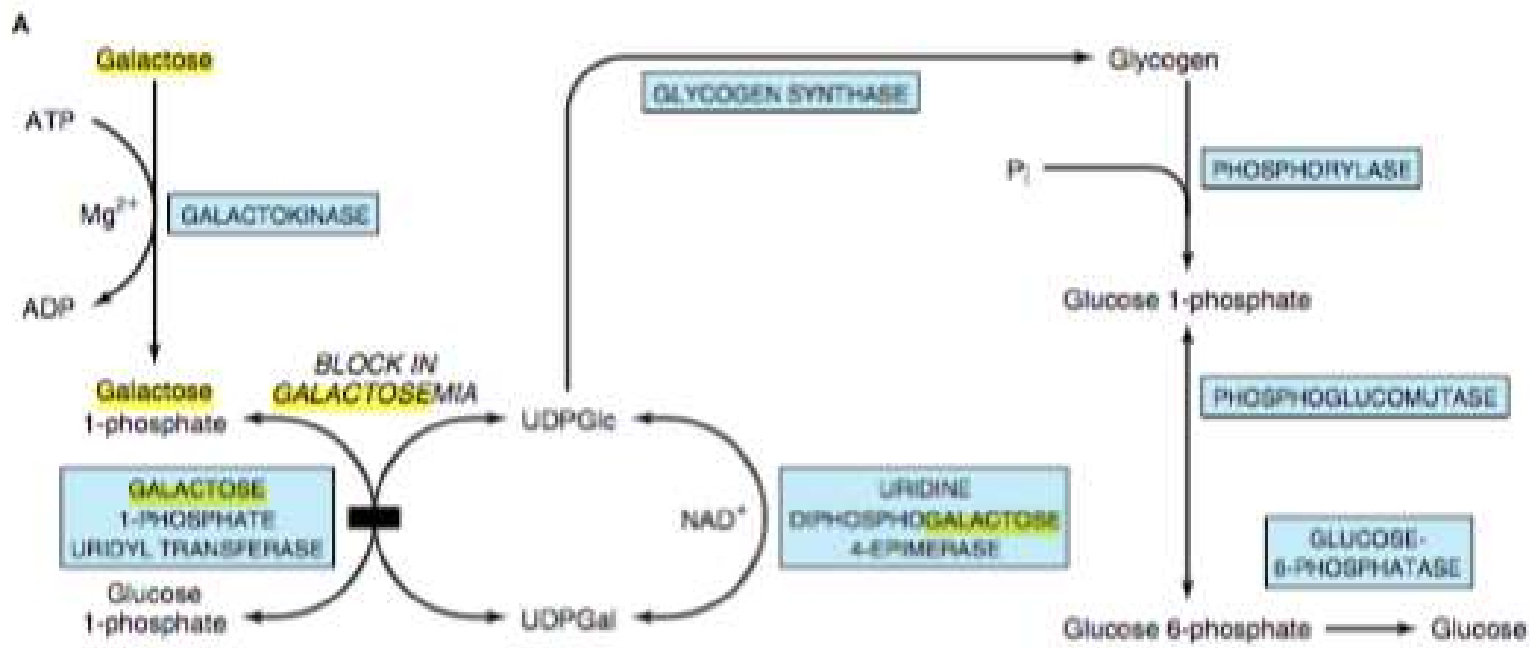
A case file: Lactose intolerance

- A patient with lactose intolerance presents with symptoms of bloating, gas, diarrhea, and abdominal pain after consuming dairy products.
- Biochemical changes:
  - Deficiency of the enzyme lactase.
  - Production of gas (hydrogen, methane).
  - Osmotic effect draws water into the colon, causing diarrhea.
- Diagnosis:
  - A lactose tolerance test
  - Genetic testing.
- Management:
  - Avoiding lactose-containing foods,
  - Using lactase supplements, or
  - Consuming lactose-free dairy products to alleviate symptoms and maintain a balanced diet.

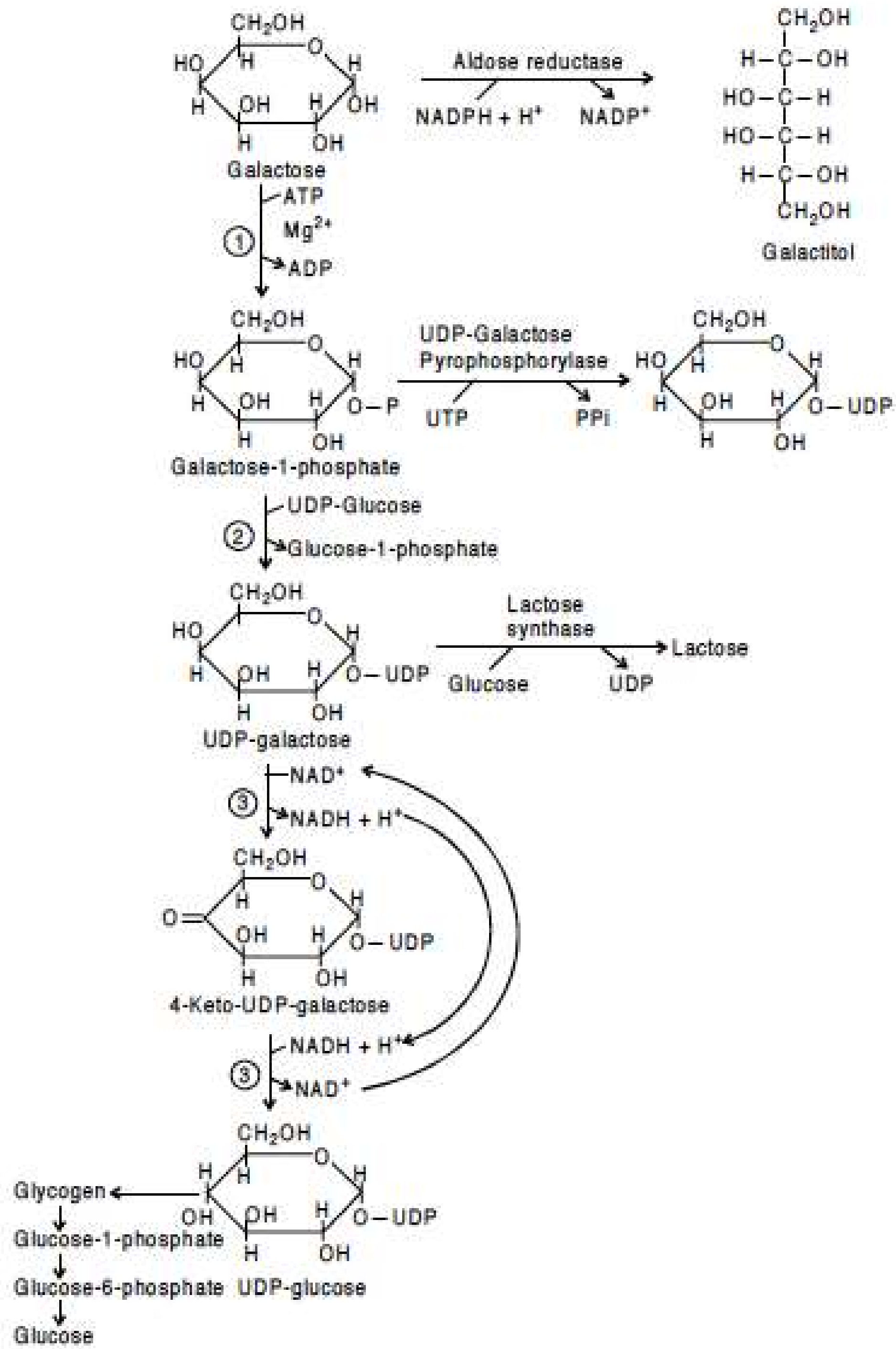
The Leloir pathway: Galactose is utilized by conversion to glucose



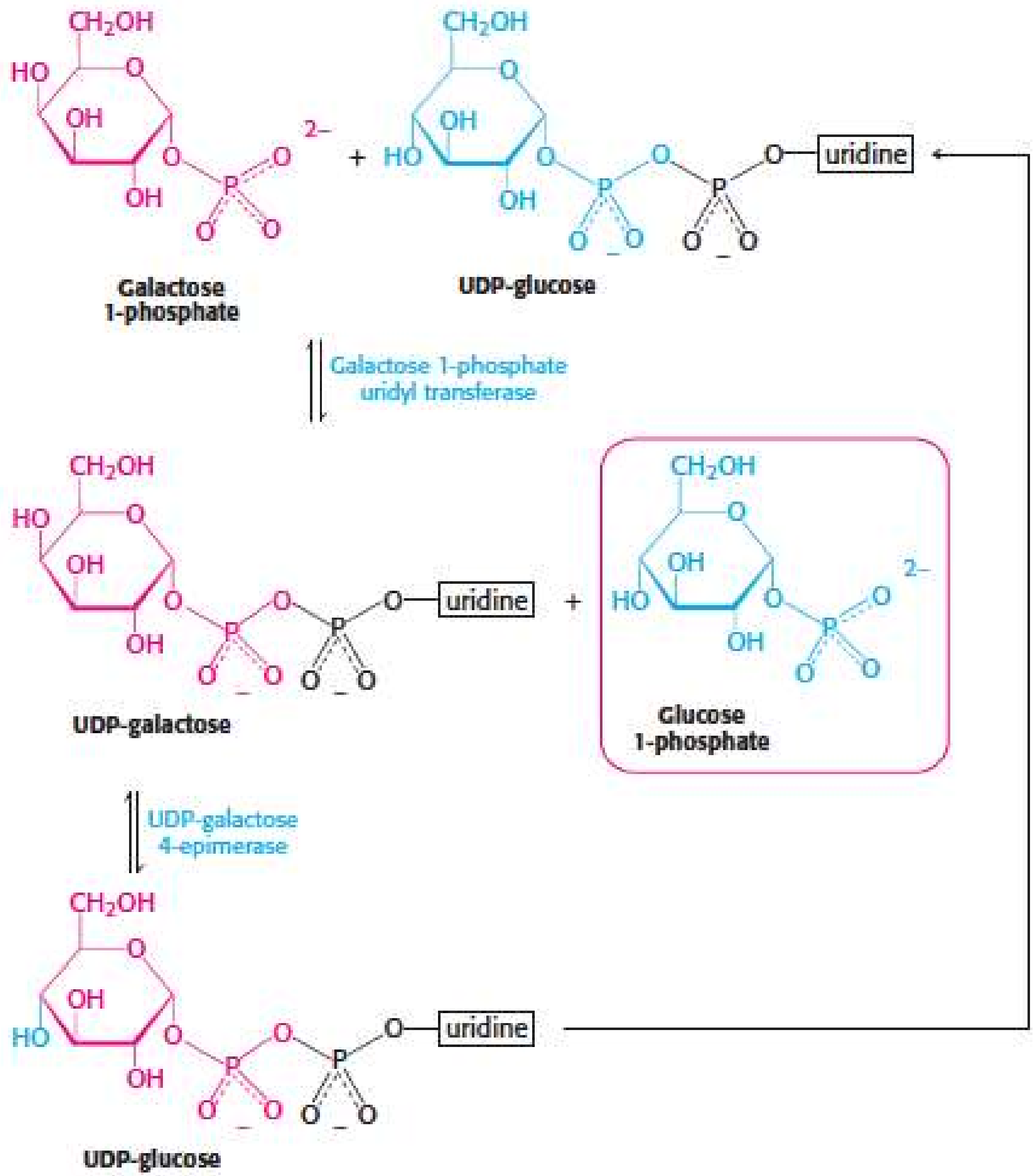
Galactose is converted to glucose-6P via a four step reaction involving UDP-glucose (<u>The Leloir pathway</u>)



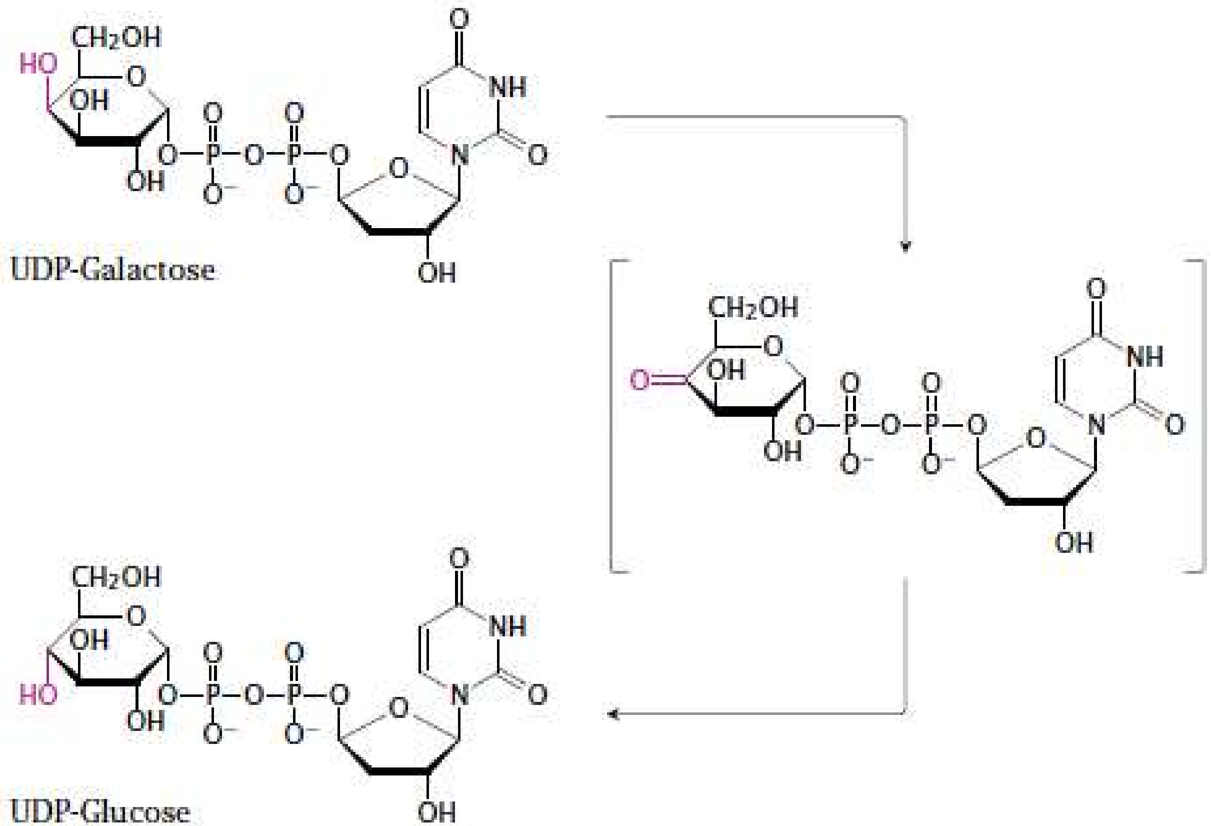
The Leloir pathway:



Galactose is utilized by conversion to glucose



Mechanism of UDP-galactose epimerase



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Galactosemia

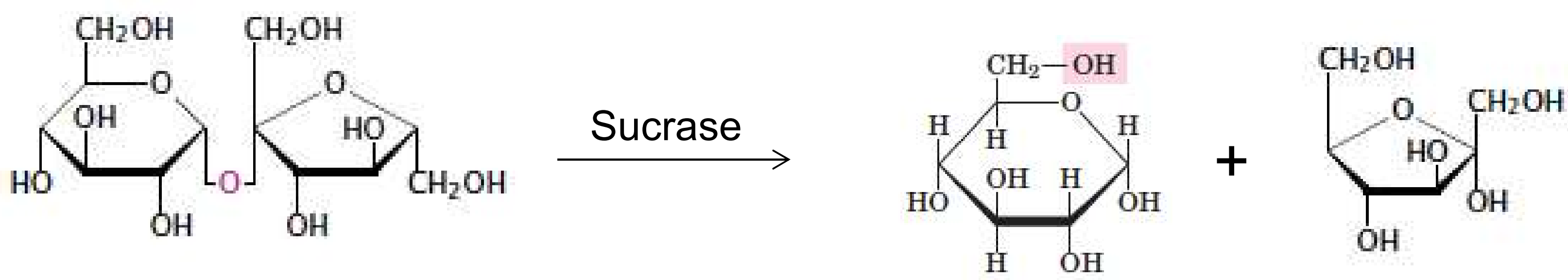
- means “galactose in the blood.”

Type	Enzyme deficiency	Accumulating metabolites
I	galactose-1-phosphate-uridyltransferase	galactose, galactose-1-phosphate, galactitol, galactonate
II	galactokinase	galactose, galactitol
III	UDP-galactose epimerase	galactose-1 phosphate, UDP-galactose

A case file: Galactosemia

- A patient with galactosemia presents with symptoms of jaundice, vomiting, and lethargy after consuming milk or dairy products.
- Biochemical changes:
  - Deficiency of the enzyme galactose-1-phosphate uridylyltransferase (GALT).
  - Accumulation of Galactose-1-phosphate, galactitol and galactonate.
- Clinical manifestations:
  - Damage to the liver, kidneys, and nervous system, leading to symptoms such as jaundice, vomiting, lethargy, and developmental delays.
- Diagnosis:
  - Newborn screening or
  - Genetic testing for GALT mutations.
- Management:
  - Avoiding milk and dairy products,
  - Using galactose-free formula, and
  - Monitoring for complications such as cataracts, developmental delays and learning disabilities.

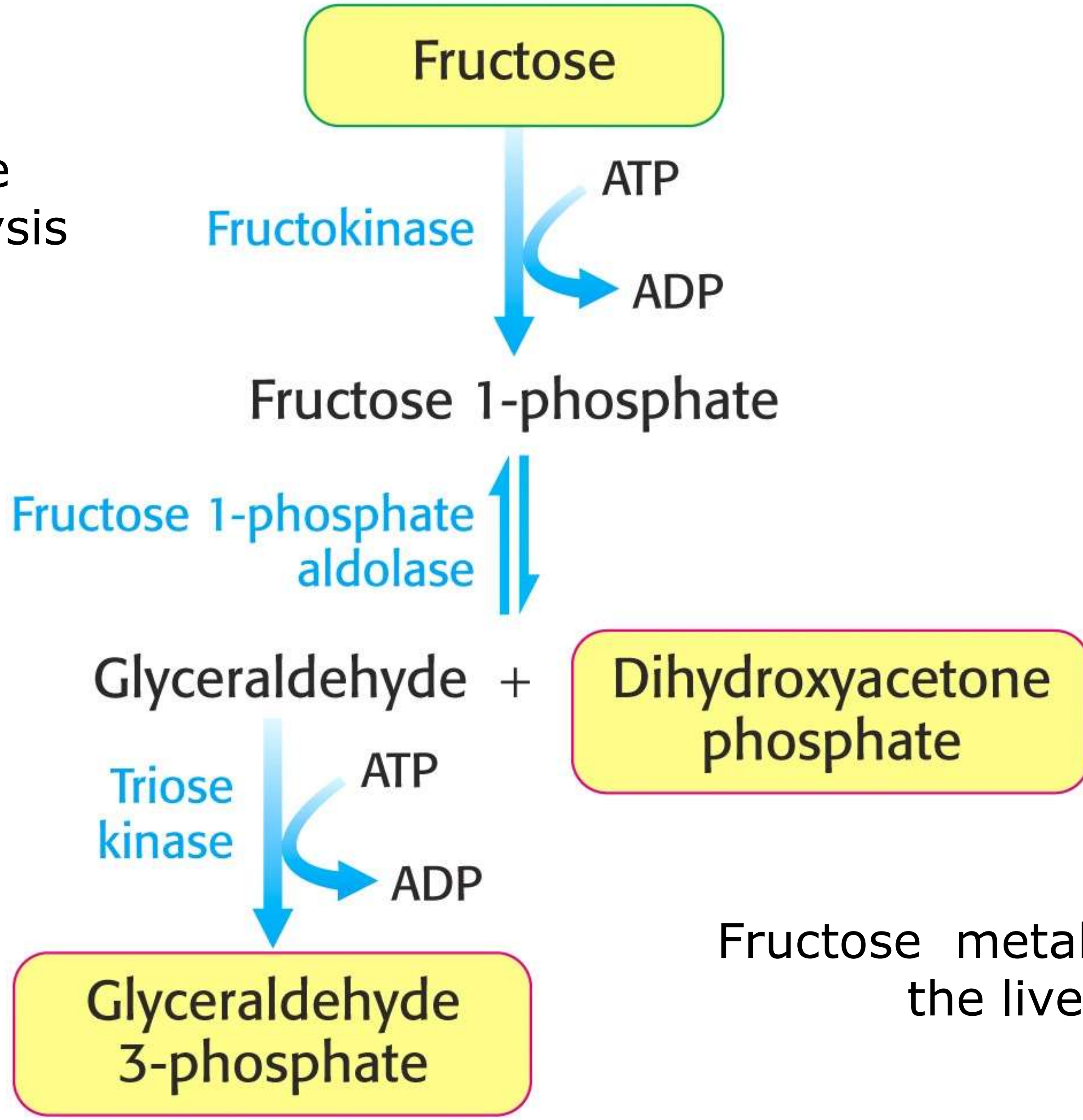
Degradation of sucrose



α-D-glucosyl-(1,2)-β-D-fructoside (sucrose) Glucose Fructose

- Sucrose consists of glucose and fructose joined by a β -glycosidic bond between the carbon 1 of glucose and carbon 2 of fructose.
- The hydrolytic cleavage of sucrose occurs at the surface of the intestinal epithelial cells using β –fructosidase (sucrase).
- Both sugars are then taken up by specific transport: Glucose by the SGLT1 transporter, and fructose by the GLUT5 transporter.

In the liver, when fructose enters glycolysis the PFK reaction is bypassed.



Fructose metabolism in the liver

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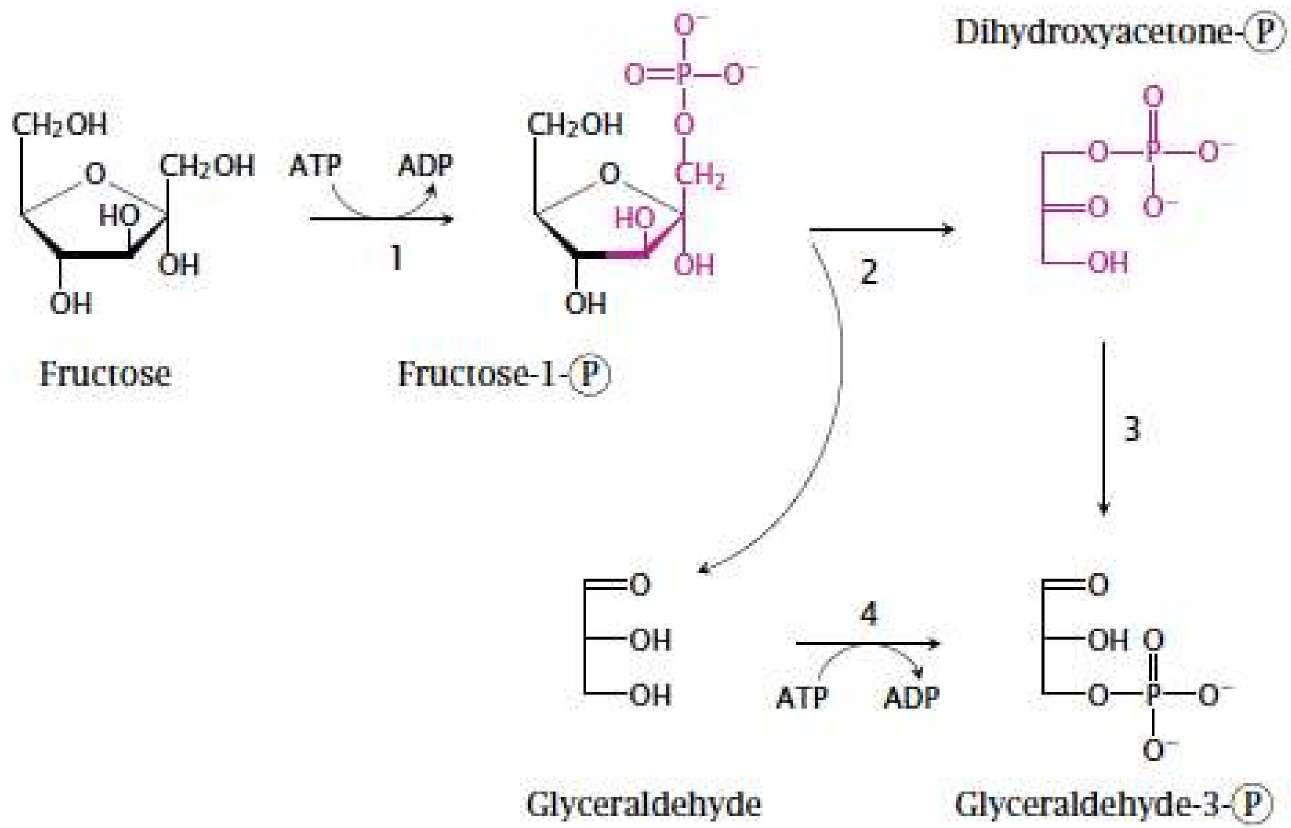
The fructolysis pathway



1=fructokinase; 2=aldolase B; 3=triose phosphate isomerase; 4=glyceraldehyde kinase

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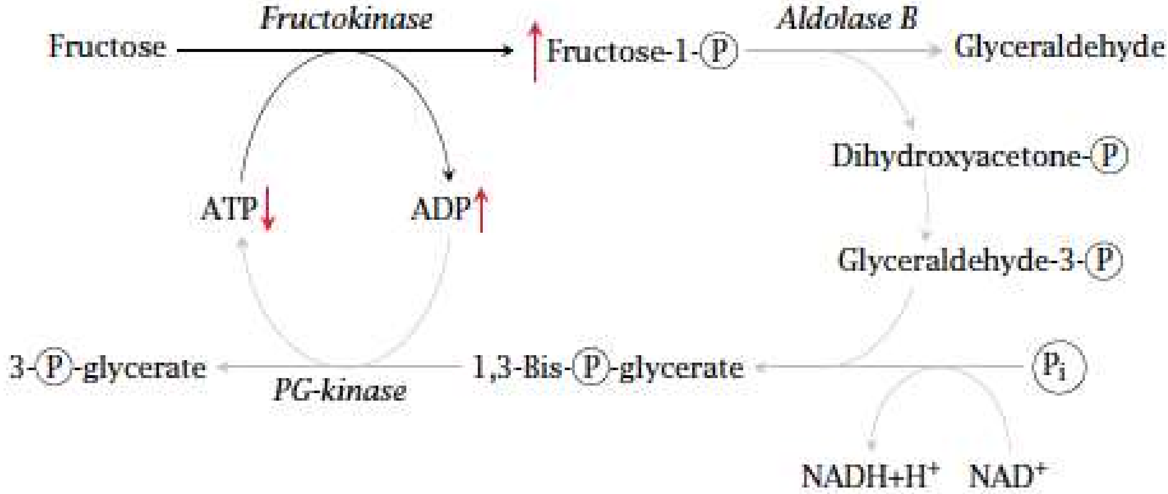
Fructose intolerance & Fructosemia



Fructose intolerance:

- a hereditary disease caused by a homozygous defect in the aldolase B gene.
- resulting in fructose-1-phosphate accumulation and lacking of ATP
- sooner or later damage or even destroy the cell and characterized by potentially severe liver failure.
- Large intravenous dosages of fructose can significantly deplete liver ATP

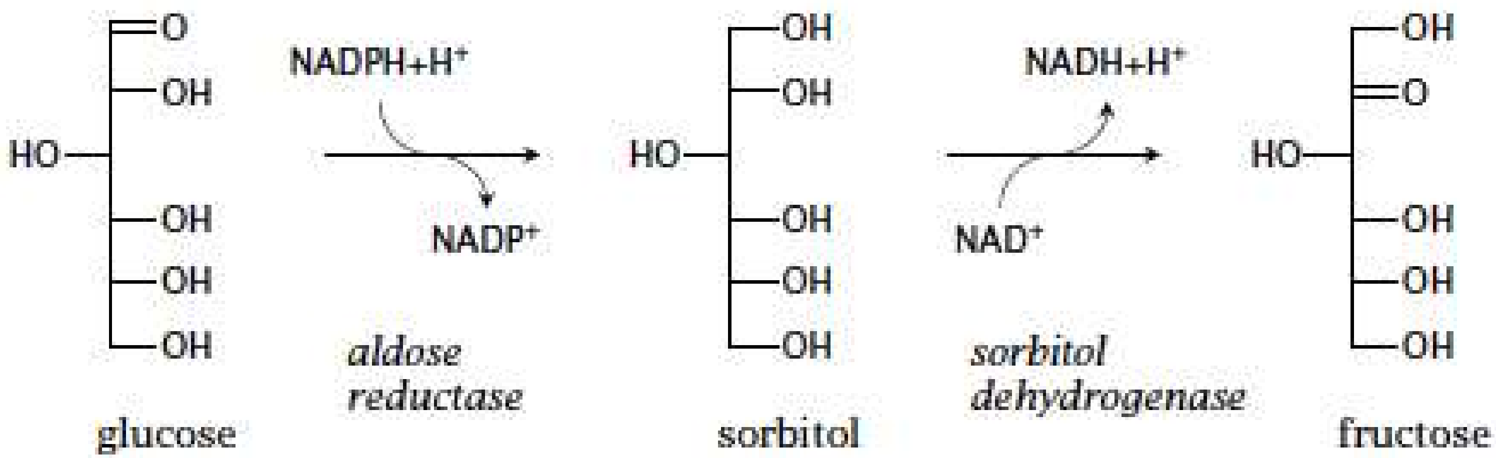
Fructosemia:

- A defect in the gene encoding fructokinase leads to a condition named fructosemia or fructosuria.
- fructose levels are increased both in the blood and the urine.
- Since fructose is not phosphorylated, no phosphate depletion occurs, and the liver cells do not incur any damage.

A case file: Fructose intolerance

- A patient with fructose intolerance presents with symptoms of abdominal pain, diarrhea, and bloating after consuming foods high in fructose, such as fruits, juices, and sweets.
- Biochemical changes:
  - Deficiency of the enzyme aldolase B.
  - Fructose-1-phosphate accumulates in the body, leading to the formation of harmful metabolites such as fructose-1,6-bisphosphate and sorbitol.
- Clinical manifestations:
  - Damage to the liver, leading to symptoms such as abdominal pain, diarrhea, and bloating.
- Diagnosis:
  - Genetic testing for aldolase B mutations or
  - A fructose tolerance test.
- Management:
  - Avoiding foods high in fructose,
  - Using low-fructose diets, and
  - Monitoring for complications such as liver damage and malnutrition.

The polyol pathway



- Aldose reductase also coverts galactose to galactitol
- Conversion of glucose to fructose via the polyol pathway occurs in the seminal vesicles (male sexual organs) and fructose is found in the sperm fluid.
- Accumulation of either sorbitol or galactitol causes cataract by cell damage through swelling.

Read more…

THANK YOU…